

restart; Trying for general coordinate system. But the equations are too complicated. Also need to remove special cases of Orthogonal basis.

```
libname := "/home/g/Asteroids/Equations", libname;
with(DAFunctions) :
libname := "/home/g/Asteroids/Equations",
"/home/g/maple2023/lib"
////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////
```

Curvilinear Coordinates:

```
with(LinearAlgebra) : x[1], x[2], x[3] := r, theta, phi; dim :=
3;
x1, x2, x3 := r, θ, φ
dim := 3
```

Relation of curvilinear coordinates with cartesian coordinates:

```
X := Vector(1..3) : X[1], X[2], X[3] := (R_E + h[2](x[2])
+ x[1])·cos(x[3])·sin(x[2]) -  $\frac{r}{R_E} \text{diff}(h[2](x[2]), x[2])$ 
·cos(x[2])·cos(x[3]), (R_E + h[2](x[2]) + r)·sin(x[3])
·sin(x[2]) -  $\frac{r}{R_E} \text{diff}(h[2](x[2]), x[2])$ ·cos(x[2])·sin(x[3]),
(R_E + h[2](x[2]) + r)·cos(x[2]) +  $\frac{r}{R_E} \text{diff}(h[2](x[2]),$ 
x[2])·sin(x[2]);
X1, X2, X3 := (R_E + h2(θ) + r) cos(φ) sin(θ)
```

$$\begin{aligned}
& - \frac{r \left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) \cos(\phi)}{R_E}, (R_E + h_2(\theta) \\
& + r) \sin(\phi) \sin(\theta) - \frac{r \left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) \sin(\phi)}{R_E}, (R_E \\
& + h_2(\theta) + r) \cos(\theta) + \frac{r \left(\frac{d}{d\theta} h_2(\theta) \right) \sin(\theta)}{R_E}
\end{aligned}$$

Defining covariant basis vectors:

$e := \text{Matrix}(1..3, 1..3) :$

for i **from** 1 **to** 3 **do**

for j **from** 1 **to** 3 **do**

$e[i, j] := \text{diff}(X[j], x[i]) :$

$e[i, j] := \text{subs}(h[2](x[2]) = \text{kappa} \cdot h[2](x[2]), e[i, j]);$

$e[i, j] := \text{convert}(\text{series}(e[i, j], \text{kappa}, 2), \text{polynom});$

$e[i, j] := \text{convert}(\text{series}(e[i, j], r, 1), \text{polynom});$

$e[i, j] := \text{subs}(\text{kappa} = 1, e[i, j]);$

end do

end do:

$e;$

$$\begin{aligned}
& \left[\left[\cos(\phi) \sin(\theta) - \frac{\left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) \cos(\phi)}{R_E}, \sin(\phi) \sin(\theta) \right. \right. \\
& \quad \left. \left. - \frac{\left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) \sin(\phi)}{R_E}, \cos(\theta) \right. \right. \\
& \quad \left. \left. + \frac{\left(\frac{d}{d\theta} h_2(\theta) \right) \sin(\theta)}{R_E} \right] \right], \\
& \left[R_E \cos(\phi) \cos(\theta) + \left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\phi) \sin(\theta) \right.
\end{aligned} \tag{4}$$

$$\begin{aligned}
& + h_2(\theta) \cos(\phi) \cos(\theta), R_E \sin(\phi) \cos(\theta) + \left(\frac{d}{d\theta} \right. \\
& \left. h_2(\theta) \right) \sin(\phi) \sin(\theta) + h_2(\theta) \sin(\phi) \cos(\theta), -R_E \sin(\theta) \\
& + \left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) - h_2(\theta) \sin(\theta) \Big], \\
& \left[\begin{aligned}
& -R_E \sin(\phi) \sin(\theta) - h_2(\theta) \sin(\phi) \sin(\theta), \\
& R_E \cos(\phi) \sin(\theta) + h_2(\theta) \cos(\phi) \sin(\theta), 0 \end{aligned} \right] \Big]
\end{aligned}$$

Metric tensor:

$M := \text{Matrix}(1 \dots 3, 1 \dots 3) :$

for i **from** 1 **to** 3 **do**

for j **from** 1 **to** 3 **do**

$M[i, j] := \text{simplify}(\text{DotProduct}(e[i], e[j], \text{conjugate} = \text{false}),$
 $\text{assume} = [0 < x[2]]);$

$M[i, j] := \text{subs}(h[2](x[2]) = \text{kappa} \cdot h[2](x[2]), M[i, j]);$

$M[i, j] := \text{convert}(\text{series}(M[i, j], \text{kappa}, 2), \text{polynom});$

$M[i, j] := \text{subs}(\text{kappa} = 1, M[i, j]);$

end do;

end do;

$M;$

$$\begin{bmatrix}
1 & 0 & 0 \\
0 & R_E^2 + 2 h_2(\theta) R_E & 0 \\
0 & 0 & \sin(\theta)^2 R_E^2 + 2 \sin(\theta)^2 h_2(\theta) R_E
\end{bmatrix} \quad (5)$$

$\text{det}_M := \text{simplify}(\text{Determinant}(M));$

$$det_M := R_E^2 (2 h_2(\theta) + R_E)^2 \sin(\theta)^2 \quad (6)$$

$U := Vector(1..3) :$

for k **from** 1 **to** 3 **do**

$U[k] := u[k](x[1], x[2], t);$

end do:

$U[1] := U[1];$

$$U_1 := u_1(r, \theta, t) \quad (7)$$

$$div_u := \left(simplify \left(add \left(\frac{1}{\sqrt{det_M}} \right. \right. \right. \\ \cdot Diff \left(\frac{U[i] \cdot \sqrt{det_M}}{\sqrt{M[i, i]}} , x[i] \right), i = 1..2 \Big), assume = [0 < r , \\ 0 < \sin(\theta), 0 < R_E, R_E + 2 \cdot h[2](x[2]) > 0] \Big) \Big);$$

$$div_u := \frac{1}{R_E (2 h_2(\theta) + R_E)} \left(\left(\frac{\partial}{\partial r} (u_1(r, \theta, t) R_E (2 h_2(\theta) \right. \right. \quad (8) \\ + R_E) \sin(\theta)) + \frac{\partial}{\partial \theta} (u_2(r, \theta, \\ t) \sqrt{R_E} \sqrt{2 h_2(\theta) + R_E} \sin(\theta)) \Big) \csc(\theta) \Big)$$

Defining christoffel symbols only for orthogonal systems:

$R := Array(1..3, 1..3, 1..3) :$

for i **from** 1 **to** 3 **do**

for j **from** 1 **to** 3 **do**

for k **from** 1 **to** 3 **do**

$$R[i, j, k] := simplify \left(DotProduct \left(diff \sim (e[i], x[j]), \right. \right. \\ \left. \left. \frac{e[k]}{M[k, k]}, conjugate = false \right) \right);$$

end do;

end do;

end do;

$R[3, 3, 1]$

$$\frac{\left(\left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) - R_E \sin(\theta) \right) (R_E + h_2(\theta)) \sin(\theta)}{R_E} \quad (9)$$

for i from 1 to 3 do

$div_uu[i] := simplify\left(simplify\left(add\left(\frac{1}{\sqrt{det_M} \cdot \sqrt{M[i, i]}} \right. \right. \right.$
 $\cdot \left(Diff\left(\frac{U[k] \cdot U[i] \cdot \sqrt{det_M} \cdot \sqrt{M[i, i]}}{\sqrt{M[k, k]}} , x[k] \right) \right), k = 1$
 $..2 \Big), assume = [0 < r, 0 < \sin(\theta)] \Big)$
 $+ simplify\left(add\left(add\left(\frac{U[k] \cdot U[j]}{\sqrt{M[k, k]} \cdot \sqrt{M[j, j]}} \cdot R[j, k, i] \right. \right. \right.$
 $\cdot \sqrt{M[i, i]}, j = 1 ..3 \Big) + \frac{2 \cdot U[k] \cdot U[i] \cdot \sqrt{M[i, i]}}{\sqrt{M[k, k]}}$
 $\cdot diff\left(\frac{1}{\sqrt{M[i, i]}} , x[k] \right), k = 1 ..3 \Big), assume = [0 < r, 0$
 $< \sin(\theta)] \Big);$

end do:

$P := Matrix(1 ..3, 1 ..3) :$

for i from 1 to 3 do

for j from 1 to 3 do

$P[i, j] := p[i, j](x[1], x[2], t);$

end do;

end do:

$P := subs(\{P[2, 1] = P[1, 2], P[3, 2] = P[2, 3], P[3, 1] = P[1,$
 $3]\}, P) :$

for i from 1 to 3 do

$$\text{div_}P[i] := \text{simplify}\left(\text{simplify}\left(\text{add}\left(\frac{1}{\text{sqrt}(\text{det_}M) \cdot \text{sqrt}(M[i, i])}\right.\right.\right. \\
\left.\left.\cdot \left(\text{Diff}\left(\frac{P[k, i] \cdot \text{sqrt}(\text{det_}M) \cdot \text{sqrt}(M[i, i])}{\text{sqrt}(M[k, k])}, x[k]\right)\right), k = 1 \dots 2\right)\right) \\
+ \text{simplify}\left(\text{add}\left(\text{add}\left(\frac{P[k, j]}{\text{sqrt}(M[k, k]) \cdot \text{sqrt}(M[j, j])} \cdot R[j, k, i]\right.\right.\right. \\
\left.\left.\cdot \text{sqrt}(M[i, i]), j = 1 \dots 3\right) + \frac{2 \cdot P[k, i] \cdot \text{sqrt}(M[i, i])}{\text{sqrt}(M[k, k])}\right. \\
\left.\cdot \text{diff}\left(\frac{1}{\text{sqrt}(M[i, i])}, x[k]\right), k = 1 \dots 3\right), \text{assume} = [0 < r, 0 \\
< \sin(\text{theta})]);$$

end do:

for k from 1 to 3 do

$$\text{div_}P[k] := \text{expand}(\text{simplify}(\text{subs}(\{P[2, 1] = P[1, 2], P[3, 2] \\
= P[2, 3], P[3, 1] = P[1, 3]\}, \text{div_}P[k])))$$

end do:

In principal CS of the body:

$$\text{Omega} := \text{Vector}(1 \dots 3) : \text{for } i \text{ from 1 to 3 do } \text{Omega}[i] := \\
\text{omega}[i](t) \text{ end do:}$$

In Body fixed curvilinear coodinates:

$$B := \text{Vector}(1 \dots 3) : \text{for } i \text{ from 1 to 3 do } B[i] := b[i](x[1], \\
x[2], x[3]) ; \text{end do:}$$

In principal CS:

$$\text{pos} := \text{Vector}(1 \dots 3) : \text{pos} := X:$$

```

Cent := simplify(CrossProduct(Omega,
  CrossProduct(Omega, pos)), assume = [0 < x[1], sin(x[2])
  > 0]) :

```

In curvilinear coordinates:

```

temp := Vector(1..3) :
for i from 1 to 3 do
temp[i] := simplify(DotProduct(Cent,  $\frac{e[i]}{\text{sqrt}(M[i, i])}$ ,
  conjugate = false), assume = [0 < x[1], sin(x[2]) > 0])
end do :
B := B - temp :

```

```

B := simplify(subs({omega[1](t) = 0, omega[2](t) = 0}, B),
  assume = [0 < x[1], sin(x[2]) > 0]) :

```

In Principal CS:

```

A := Vector(1..3) : for i from 1 to 3 do A[i] := alpha[i](t);
end do :

```

```

A_acc := CrossProduct(A, pos) :

```

In CCS:

```

temp[1..3] := A_acc[1..3] :
for i from 1 to 3 do
A_acc[i] := simplify(DotProduct(temp,  $\frac{e[i]}{\text{sqrt}(M[i, i])}$ ,
  conjugate = false), assume = [0 < x[1], sin(x[2]) > 0]);
end do :

```

```

A_acc := simplify(subs({alpha[1](t) = 0, alpha[2](t) = 0},
  A_acc), assume = [0 < x[1], sin(x[2]) > 0]);

```

$$A_{acc} := \begin{bmatrix} 0 \end{bmatrix},$$

(10)

$$\begin{bmatrix} 0 \\ \left[\frac{1}{\sqrt{R_E (2 h_2(\theta) + R_E)} R_E} \left((R_E + h_2(\theta)) \left(-\left(\frac{d}{d\theta} h_2(\theta) \right) r \cos(\theta) + R_E \sin(\theta) (R_E + h_2(\theta) + r) \right) \alpha_3(t) \right) \right] \end{bmatrix}$$

In BFCS:

$\Omega_B := \text{Vector}(1..3) :$

for i **from** 1 **to** 3 **do**

$\Omega_B[i] := \text{simplify}\left(\text{DotProduct}\left(\Omega, \frac{e[i]}{\text{sqrt}(M[i, i])}, \right. \right.$
 $\left. \text{conjugate} = \text{false} \right), \text{assume} = [0 < x[1], \sin(x[2]) > 0] \Big);$

end do;

$\text{corr} := \text{simplify}(\text{CrossProduct}(\Omega_B, U), \text{assume} = [0 < x[1], \sin(x[2]) > 0]) :$

$\text{corr} := \text{simplify}(\text{subs}(\{\omega[1](t) = 0, \omega[2](t) = 0\}, \text{corr}), \text{assume} = [0 < x[1]]);$

$$\text{corr} := \left[\left[\frac{1}{\sqrt{R_E (2 h_2(\theta) + R_E)}} \left(\left(\left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) - \sin(\theta) (R_E + h_2(\theta)) \right) \omega_3(t) u_3(r, \theta, t) \right) \right] \right], \quad (11)$$

$$\left[-\frac{u_3(r, \theta, t) \omega_3(t) \left(\left(\frac{d}{d\theta} h_2(\theta) \right) \sin(\theta) + R_E \cos(\theta) \right)}{R_E} \right],$$

$$\left[\frac{1}{\sqrt{R_E (2 h_2(\theta) + R_E)} R_E} \left(\omega_3(t) \left(u_2(r, \theta, t) \left(\left(\frac{d}{d\theta} h_2(\theta) \right) \sin(\theta) + R_E \cos(\theta) \right) \sqrt{R_E (2 h_2(\theta) + R_E)} \right. \right. \right. \\ \left. \left. \left. - \left(\left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) - \sin(\theta) (R_E + h_2(\theta)) \right) R_E u_1(r, \theta, t) \right) \right) \right]$$

Acceleration of the core in PCS:

$A_core := Vector(1..3) :$

for i **from** 1 **to** 3 **do**

$A_core[i] := a[i](t)$

end do;

Acceleration of the core in CCS:

$a_B := Vector(1..3) :$

for i **from** 1 **to** 3 **do**

$a_B[i] := simplify \left(DotProduct \left(A_core, \frac{e[i]}{\sqrt{M[i, i]}}, \right. \right. \\ \left. \left. conjugate = false \right), assume = [0 < x[1], \sin(x[2]) > 0] \right)$

end do;

$$a_B_1 := \frac{1}{R_E} \left(\left((-a_2(t) \sin(\phi) - \cos(\phi) a_1(t)) \cos(\theta) \right. \right. \\ \left. \left. + a_3(t) \sin(\theta) \right) \left(\frac{d}{d\theta} h_2(\theta) \right) + R_E (\cos(\theta) a_3(t) \right. \\ \left. + \sin(\theta) (a_2(t) \sin(\phi) + \cos(\phi) a_1(t)) \right)$$

$$a_B_2 := \frac{1}{\sqrt{R_E (2 h_2(\theta) + R_E)}} \left((\cos(\theta) a_3(t) \right.$$

$$\begin{aligned}
& + \sin(\theta) \left(a_2(t) \sin(\phi) + \cos(\phi) a_1(t) \right) \left(\frac{d}{d\theta} h_2(\theta) \right) \\
& + \left(R_E + h_2(\theta) \right) \left(\left(a_2(t) \sin(\phi) + \cos(\phi) a_1(t) \right) \cos(\theta) \right. \\
& \left. - a_3(t) \sin(\theta) \right) \\
a_{-B_3} &:= \frac{\left(R_E + h_2(\theta) \right) \left(a_2(t) \cos(\phi) - a_1(t) \sin(\phi) \right)}{\sqrt{R_E \left(2 h_2(\theta) + R_E \right)}} \tag{12}
\end{aligned}$$

Momentum balance:

mom := *Vector*(1..3) :

for *i* **from** 1 **to** 3 **do**

div_uu[*i*] := *convert*(*div_uu*[*i*], *sincos*);

div_P[*i*] := *convert*(*div_P*[*i*], *sincos*);

mom[*i*] := *rho*·*Diff*(*U*[*i*], *t*) + *rho*·*div_uu*[*i*] + *div_P*[*i*] − *rho*
·*bf*[*i*](*x*[1], *x*[2], *x*[3], *t*);

mom[*i*] := *subs*(*h*[2](*x*[2]) = *kappa*·*h*[2](*x*[2]), *mom*[*i*]);

mom[*i*] := *convert*(*series*(*mom*[*i*], *kappa*, 2), *polynom*);

mom[*i*] := *simplify*(*subs*(*kappa* = 1, *mom*[*i*]), *assume* = [0
< *r*, 0 < *sin*(*theta*), 0 < *R_E*, *R_E* + 2· *h*[2](*x*[2]) > 0]);

end do:#*BF* also contains acceleration of the core

cont := *numer*(*convert*(*div_u*, *sincos*));

mom[1];

$$cont := \frac{\partial}{\partial r} \left(u_1(r, \theta, t) R_E \left(2 h_2(\theta) + R_E \right) \sin(\theta) \right) + \frac{\partial}{\partial \theta}$$

$$\left(u_2(r, \theta, t) \sqrt{R_E} \sqrt{2 h_2(\theta) + R_E} \sin(\theta) \right)$$

$$\frac{1}{R_E^3} \left(-2 \rho \csc(\theta) \left(h_2(\theta) - \frac{R_E}{2} \right) \left(\frac{\partial}{\partial r} \left(u_1(r, \theta, \right. \right. \tag{13}$$

$$t)^2 \sin(\theta) R_E^2 \right) + \left(\frac{\partial}{\partial r} \left(2 u_1(r, \theta,$$

$$t)^2 \sin(\theta) R_E h_2(\theta) \right) \Big) R_E \rho \csc(\theta) - 2 \csc(\theta) \left(h_2(\theta)$$

$$\begin{aligned}
& - \frac{R_E}{2} \left(\frac{\partial}{\partial r} \left(p_{1,1}(r, \theta, t) \sin(\theta) R_E^2 \right) \right) + \left(\frac{\partial}{\partial \theta} \left(u_2(r, \theta, \right. \right. \\
& \left. \left. t) u_1(r, \theta, t) \sin(\theta) h_2(\theta) \right) \right) R_E \rho \csc(\theta) \\
& - 2 \rho \csc(\theta) \left(h_2(\theta) - \frac{R_E}{2} \right) \left(\frac{\partial}{\partial \theta} \left(u_2(r, \theta, t) u_1(r, \theta, \right. \right. \\
& \left. \left. t) \sin(\theta) R_E \right) \right) + \left(\frac{\partial}{\partial r} \left(2 p_{1,1}(r, \theta, \right. \right. \\
& \left. \left. t) \sin(\theta) R_E h_2(\theta) \right) \right) R_E \csc(\theta) + \left(\frac{\partial}{\partial \theta} \left(p_{1,2}(r, \theta, \right. \right. \\
& \left. \left. t) \sin(\theta) h_2(\theta) \right) \right) R_E \csc(\theta) - 2 \csc(\theta) \left(h_2(\theta) \right. \\
& \left. - \frac{R_E}{2} \right) \left(\frac{\partial}{\partial \theta} \left(p_{1,2}(r, \theta, t) \sin(\theta) R_E \right) \right) + \left(\left(u_2(r, \theta, t) \right)^2 \rho \right. \\
& \left. + p_{2,2}(r, \theta, t) \right) \left(\frac{d^2}{d\theta^2} h_2(\theta) \right) + \left(\frac{\partial}{\partial t} u_1(r, \theta, t) \right) R_E^2 \rho \\
& - b f_1(r, \theta, \phi, t) R_E^2 \rho + (h_2(\theta) - R_E) p_{2,2}(r, \theta, t) \\
& + \left(\left(\frac{d}{d\theta} h_2(\theta) \right) \cot(\theta) + h_2(\theta) - R_E \right) p_{3,3}(r, \theta, t) \\
& + \rho \left((h_2(\theta) - R_E) u_2(r, \theta, t)^2 + u_3(r, \theta, t)^2 \left(\left(\frac{d}{d\theta} \right. \right. \right. \\
& \left. \left. h_2(\theta) \right) \cot(\theta) + h_2(\theta) - R_E \right) \left. \right) \left. \right) R_E
\end{aligned}$$

for i from 1 to 3 do

$mom[i] := collect(mom[i], rho)$

end do:

Boundary conditions:

$H := Vector(1..2) :$

$H[1] := h[1](x[2], t) :$

$H[2] := 0 :$

$K_BC := Vector(1..2) :$

$N := Vector(1..2) :$

for i from 1 to 2 do

$N[i] := subs\left(x[1] = H[i], convert\left(\left[\frac{diff(x[1] - H[i], x[1])}{sqrt(M[1, 1])}, \frac{diff(x[1] - H[i], x[2])}{sqrt(M[2, 2])}, \frac{diff(x[1] - H[i], x[3])}{sqrt(M[3, 3])}\right], Vector\right)\right);$

if i = 1 then $K_BC[i] := simplify(subs((x[1] = H[i]), DotProduct(U, N[i], conjugate = false) = diff(H[i], t)), assume = [H[i] > 0, sin(x[2]) > 0]);$

else $K_BC[i] := simplify(subs((x[1] = H[2]), DotProduct(U, N[i], conjugate = false) = diff(H[i], t)), assume = [H[i] > 0, sin(x[2]) > 0]);$

end if

end do:

$K_BC; N[1];$

$$\left[\left[\frac{1}{\sqrt{R_E (2 h_2(\theta) + R_E)}} \left(u_1(h_1(\theta, t), \theta, t) \sqrt{R_E (2 h_2(\theta) + R_E)} - u_2(h_1(\theta, t), \theta, t) \left(\frac{\partial}{\partial \theta} h_1(\theta, t) \right) \right) \right] = \frac{\partial}{\partial t} h_1(\theta, t) \right],$$

$$\left[u_1(0, \theta, t) = 0 \right]$$

$$\begin{bmatrix} 1 \\ -\frac{\frac{\partial}{\partial \theta} h_1(\theta, t)}{\sqrt{R_E^2 + 2 h_2(\theta) R_E}} \\ 0 \end{bmatrix} \quad (14)$$

```

n_cont := Call_Diff(simplify(nondim(cont), assume = [0 < r ,
    0 < sin(theta), 0 < G, 0 < R_E, R_E + 2· h[2](x[2]) > 0])) :
n_cont := simplify(selection( simplify(n_cont, assume = [0
    < r , 0 < sin(theta), 0 < R_E, R_E + 2· h[2](x[2]) > 0])))) :
n_cont := convert(series(n_cont, kappa, 2), polynom) :
n_cont := simplify(convert(series(n_cont, epsilon, 1),
    polynom), assume = [0 < sin(x[2])]);
n_cont :=  $\sqrt{G} \left( \kappa \left( \frac{\partial}{\partial \theta} (u_2(r, \theta, t) h_2(\theta) \sin(\theta)) \right) + \kappa \left( \frac{\partial}{\partial r} \right. \right. \quad (15)$ 
```

$$\left. \left(2 u_1(r, \theta, t) h_2(\theta) \sin(\theta) \right) + \frac{\partial}{\partial \theta} (u_2(r, \theta, t) \sin(\theta)) + \frac{\partial}{\partial r} (u_1(r, \theta, t) \sin(\theta)) \right)$$

n_mom := Vector(1..3) :

for *i* **from** 1 **to** 3 **do**

n_mom[*i*] := nondim(mom[*i*]);

n_mom[*i*] := simplify(selection(simplify(*n_mom*[*i*]) , assume
 = [0 < *r* , 0 < sin(theta), 0 < *G*, 0 < *R_E*, *R_E* + 2
 · *h*[2](*x*[2]) > 0]));

n_mom[*i*] := convert(series(*n_mom*[*i*], kappa, 2), polynom) :

n_mom[*i*] := convert(series(*n_mom*[*i*], epsilon, 2), polynom);

end do:

simplify((subs({Diff = diff, epsilon = 0}, *n_mom*[1])), assume
 = [0 < sin(*x*[2]), 0 < *R_E*, epsilon·*x*[1] + 1 > 0]);

$$G \left(\kappa u_2(r, \theta, t)^2 \left(\frac{d^2}{d\theta^2} h_2(\theta) \right) + \frac{\partial}{\partial r} p_{1,1}(r, \theta, t) - bf_1(r, \theta, \phi, \quad (16) \right.$$

$$t) + (h_2(\theta) \kappa - 1) u_2(r, \theta, t)^2 + \left(\kappa \cot(\theta) \left(\frac{d}{d\theta} h_2(\theta) \right) \right. \\ \left. + h_2(\theta) \kappa - 1 \right) u_3(r, \theta, t)^2 \Bigg)$$

$n_K_BC := \text{Vector}(1..2) :$

for i **from** 1 **to** 2 **do**

$n_K_BC[i] := \text{simplify}(\text{nondim}(K_BC[i]), \text{assume} = [0 < R_E,$
 $0 < G, 0 < \text{epsilon}]);$

$n_K_BC[i] := \text{simplify}(\text{convert}(\text{selection}(n_K_BC[i])$
 $\cdot \sin(x[2]), \text{sincos}));$

$\#n_K_BC[i] := \text{simplify}(\text{convert}(\text{series}(\text{op}(n_K_BC[i], \text{kappa},$
 $2), \text{polynom}));$

end do:

$n_K_BC;$

$$\left[\left[\frac{1}{\sqrt{2 h_2(\theta) \kappa + 1}} \left(\sin(\theta) \sqrt{G} \left(u_1(h_1(\theta, t), \theta, \right. \right. \right. \quad (17)$$

$$t) \sqrt{2 h_2(\theta) \kappa + 1} - u_2(h_1(\theta, t), \theta, t) \left(\frac{\partial}{\partial \theta} h_1(\theta, t) \right) \Bigg) \right] \\ = \sin(\theta) \sqrt{G} \left(\frac{\partial}{\partial t} h_1(\theta, t) \right) \Bigg],$$

$$\left[\sin(\theta) e u_1(0, \theta, t) = 0 \right]$$

$D_BC := \text{Vector}(1..2) :$

$D_BC[1] := \text{subs}\left(x[1] = H[1],$

$\text{simplify}\left(\frac{P.N[1]}{\text{VectorCalculus[Norm]}(N[1])}, \text{assume} = [0$
 $< x[1], \sin(x[2]) > 0]\right) \Bigg) :$

$D_BC[2] := \text{subs}\left(x[1] = H[2], \right.$
 $\left. \text{simplify}\left(\frac{P.N[2]}{\text{VectorCalculus[Norm]}(N[2])}, \text{assume} = [0\right.$
 $\left. < x[1], \sin(x[2]) > 0]\right)\right):$

for i from 1 to 3 do

$D_BC[1][i] := \text{simplify}(D_BC[1][i] = 0, \text{assume} = [0 < x[1],$
 $\sin(x[2]) > 0]);$

$D_BC[2][i] := \text{simplify}(D_BC[2][i] = Pb[i](x[2], x[3], t)$
 $+ Fr[i](x[2], x[3], t), \text{assume} = [0 < x[1], \sin(x[2]) > 0]);$

end do:

$D_BC[2];$

$$\left[\begin{array}{l} p_{1,1}(0, \theta, t) = Pb_1(\theta, \phi, t) + Fr_1(\theta, \phi, t) \\ p_{1,2}(0, \theta, t) = Pb_2(\theta, \phi, t) + Fr_2(\theta, \phi, t) \\ p_{1,3}(0, \theta, t) = Pb_3(\theta, \phi, t) + Fr_3(\theta, \phi, t) \end{array} \right] \quad (18)$$

$n_D_BC := \text{Vector}(1..2):$

for i from 1 to 2 do

for j from 1 to 3 do

$n_D_BC[i][j] := \text{nondim}(D_BC[i][j]);$

$n_D_BC[i][j] := \text{selection}(\text{simplify}(n_D_BC[i][j], \text{assume} = [0$
 $< R_E, 0 < G, \sin(x[2]) > 0, \text{epsilon} > 0]))):$

$n_D_BC[i][j] := \text{Linearize}(\text{convert}(\text{subs}(\text{diff} = \text{Diff},$
 $\text{expand}(\text{subs}(\text{Diff} = \text{diff}, \text{simplify}(\text{selection}(n_D_BC[i][j])$
 $\cdot \sin(x[2]), \text{assume} = [0 < R_E, 0 < G, 0 < \text{epsilon}, \sin(x[2])$
 $> 0])))), \text{sincos}), \text{epsilon}, 1);$

$n_D_BC[i][j] := \text{expand}\left(\text{subs}\left(\left\{x[1] = H[i], \right.\right.\right.$

$$\left(\kappa^2 h_2(\theta, \phi)^2 \sin(\theta)^2 + \kappa^2 (\text{Diff}(h_2(\theta, \phi), \theta))^2 \sin(\theta)^2 \right.$$

$$\left. + \kappa^2 (\text{Diff}(h_2(\theta, \phi), \phi))^2 + 2 \kappa h_2(\theta, \phi) \sin(\theta)^2 \right)$$

$$\left. + \sin(\theta)^2 \right)^{1/2} = K(x[2], x[3]) \Big\}, n_D_BC[i][j] \Big) \Big) :$$

end do:

end do:

$n_D_BC[2][2];$

$$\sin(\theta) \delta p_{1,2}(0, \theta, t) = \sin(\theta) Fr_2(\theta, \phi, t) \delta + \sin(\theta) Pb_2(\theta, \phi, t) \quad (19)$$

$Dev := Matrix(1..3, 1..3) :$

for i from 1 to 3 do

for j from 1 to 3 do $Dev[i, j] :=$

$$\begin{aligned} & \text{simplify} \left(\frac{\text{diff}(\text{sqrt}(M[i, i]) \cdot U[i], x[j])}{\text{sqrt}(M[i, i] \cdot M[j, j])} \right. \\ & \quad \left. - \frac{\text{add}(R[i, j, k] \cdot \text{sqrt}(M[k, k]) \cdot U[k], k = 1..3)}{\text{sqrt}(M[i, i] \cdot M[j, j])}, \text{assume} \right. \\ & \quad \left. = [x[1] > 0, \sin(x[2]) > 0] \right); \end{aligned}$$

end do:

end do:

$$Dev := \frac{1}{2} \cdot (Dev + LinearAlgebra[Transpose](Dev)) :$$

$$\begin{aligned} Dev := & \text{simplify} \left(\frac{Dev}{\text{sqrt} \left(\frac{1}{2} \cdot LinearAlgebra[Trace](Dev.Dev) \right)}, \right. \\ & \left. \text{assume} = [x[1] > 0, \sin(x[2]) > 0] \right) : \end{aligned}$$

for i from 1 to 3 do

for j from 1 to 3 do

$Dev[i, j] := \text{simplify}(\text{nondim}(Dev[i, j]), \text{assume} = [0 < R_E, 0 < G, \sin(x[2]) > 0, \text{epsilon} > 0, \text{kappa} > 0]);$

$Dev[i, j] := \text{simplify}(\text{convert}(\text{series}(Dev[i, j], \text{kappa}, 2), \text{polynom}), \text{assume} = [\sin(x[2]) > 0, \text{kappa} > 0, \text{lambda}(x[2], x[3]) > 0]);$

$Dev[i, j] := \text{simplify}(\text{convert}(\text{series}(Dev[i, j], \text{epsilon}, 1), \text{polynom}), \text{assume} = [\sin(x[2]) > 0, \text{kappa} > 0]);$

end do: end do:

Dev;

$$\left[\begin{array}{c} 0, \frac{\frac{\partial}{\partial r} u_2(r, \theta, t)}{\sqrt{\left(\frac{\partial}{\partial r} u_2(r, \theta, t)\right)^2 + \left(\frac{\partial}{\partial r} u_3(r, \theta, t)\right)^2}}, \\ \frac{\frac{\partial}{\partial r} u_3(r, \theta, t)}{\sqrt{\left(\frac{\partial}{\partial r} u_2(r, \theta, t)\right)^2 + \left(\frac{\partial}{\partial r} u_3(r, \theta, t)\right)^2}} \end{array} \right], \quad (20)$$

$$\left[\begin{array}{c} \frac{\frac{\partial}{\partial r} u_2(r, \theta, t)}{\sqrt{\left(\frac{\partial}{\partial r} u_2(r, \theta, t)\right)^2 + \left(\frac{\partial}{\partial r} u_3(r, \theta, t)\right)^2}}, 0, 0 \end{array} \right],$$

$$\left[\begin{array}{c} \frac{\frac{\partial}{\partial r} u_3(r, \theta, t)}{\sqrt{\left(\frac{\partial}{\partial r} u_2(r, \theta, t)\right)^2 + \left(\frac{\partial}{\partial r} u_3(r, \theta, t)\right)^2}}, 0, 0 \end{array} \right]$$

for i from 1 to 3 do

$$Fr[i] := simplify \left(- \left(U[i] \right. \right. \\ \left. \left. - \frac{1}{VectorCalculus[Norm](N[2])} \left(DotProduct \left(U, \right. \right. \right. \\ \left. \left. \frac{N[2]}{VectorCalculus[Norm](N[2])}, conjugate = false \right) \cdot N[2][i] \right) \right) \\ \left. \right) / \left(VectorCalculus[Norm] \left(U \right. \right.$$

$$\begin{aligned}
& - \frac{1}{\text{VectorCalculus}[\text{Norm}](N[2])} \left(\text{DotProduct} \left(U, \right. \right. \\
& \left. \left. \frac{N[2]}{\text{VectorCalculus}[\text{Norm}](N[2])}, \text{conjugate} = \text{false} \right) \cdot N[2] \right) \Big) \Big) \\
& \cdot \frac{\text{DotProduct}(P.N[2], N[2], \text{conjugate} = \text{false})}{\text{VectorCalculus}[\text{Norm}](N[2])^2} \cdot \text{delta}, \\
& \text{assume} = [0 < x[1], \sin(x[2]) > 0, H[1] > 0, H[2] > 0] \Big);
\end{aligned}$$

$$\begin{aligned}
Pb[i] := & \\
& \text{simplify} \left(\frac{1}{\text{VectorCalculus}[\text{Norm}](N[2])^3} (\text{DotProduct}(P \right. \\
& \cdot N[2], N[2], \text{conjugate} = \text{false}) N[2][i]), \text{assume} = [0 < x[1], \\
& \left. \sin(x[2]) > 0, H[2] > 0, H[1] > 0] \right);
\end{aligned}$$

$$\begin{aligned}
BF[i] := & B[i] - 2 \cdot \text{corr}[i] - A_acc[i]; \\
\mathbf{end\ do:}
\end{aligned}$$

$$BF[2];$$

$$\begin{aligned}
& - \frac{1}{\sqrt{R_E (2 h_2(\theta) + R_E)} R_E} \left(-b_2(r, \theta, \right. \\
& \phi) \sqrt{R_E (2 h_2(\theta) + R_E)} R_E + \left(r \left(\frac{d}{d\theta} \right. \right. \\
& \left. \left. h_2(\theta) \right)^2 \cos(\theta) \sin(\theta) + \left(2 r \left(R_E + \frac{h_2(\theta)}{2} \right) \cos(\theta) \right)^2 \right. \\
& \left. - R_E \left((R_E + h_2(\theta)) \sin(\theta)^2 + r \right) \right) \left(\frac{d}{d\theta} h_2(\theta) \right) \\
& \left. - R_E \cos(\theta) \sin(\theta) (R_E + h_2(\theta)) (R_E + h_2(\theta) + r) \right)
\end{aligned} \tag{21}$$

$$\omega_3(t)^2 \Bigg) + \frac{2 u_3(r, \theta, t) \omega_3(t) \left(\left(\frac{d}{d\theta} h_2(\theta) \right) \sin(\theta) + R_E \cos(\theta) \right)}{R_E}$$

for j from 1 to 3 do

```

Fr[j] := simplify(  $\frac{\text{nondim}(Fr[j])}{\text{delta} \cdot \text{epsilon} \cdot G \cdot R\_E \cdot \text{rho}}$  ):
Fr[j] := simplify(convert(series(Fr[j], epsilon, 1), polynom),
  assume = [0 < R_E, 0 < G, sin(x[2]) > 0, kappa > 0]):
Fr[j] := simplify(convert(series(Fr[j], kappa, 2), polynom),
  assume = [0 < R_E, 0 < G, sin(x[2]) > 0, kappa > 0]):

Fr[j] := simplify(collect(subs(x[1] = H[2], Fr[j]), epsilon),
  assume = [0 < R_E, 0 < G, sin(x[2]) > 0, kappa > 0]):

Pb[j] := simplify(  $\frac{\text{nondim}(Pb[j])}{\text{epsilon} \cdot G \cdot R\_E \cdot \text{rho}}$  ):
Pb[j] := simplify(convert(series(Pb[j], epsilon, 2),
  polynom)):
Pb[j] := simplify(convert(series(Pb[j], kappa, 2), polynom),
  assume = [0 < R_E, 0 < G, sin(x[2]) > 0, kappa > 0]):

Pb[j] := simplify(collect(subs(x[1] = H[2], Pb[j]), epsilon),
  assume = [0 < R_E, 0 < G, sin(x[2]) > 0, kappa > 0]):
BF[j] := simplify( nondim( BF[j]), assume = [H[1] - H[2] > 0,
  R_E > 0, G > 0, kappa > 0]):
BF[j] := simplify(convert(series(BF[j], epsilon, 1),
  polynom));
BF[j] := simplify(convert(series(BF[j], kappa, 2), polynom),
  assume = [0 < R_E, 0 < G, sin(x[2]) > 0, kappa > 0]):
#BF[j] := local_integrate( BF[j] ): BF[j] :=
  simplify(  $\frac{\text{Final\_form}(BF[j])}{G}$ , assume = [H[1] - H[2] > 0,
    R_E > 0, G > 0]):
#BF[j] := simplify(BF[j], assume = [0 < R_E, G > 0]):

```

end do:

$BF[2]; Fr[2]; Pb[2];$

$$\begin{aligned}
 G \left(\omega_3(t) \kappa \sin(\theta) \left(\omega_3(t) \sin(\theta) + 2 u_3(r, \theta, t) \right) \left(\frac{d}{d\theta} h_2(\theta) \right) \right. \\
 \left. + 2 \omega_3(t) u_3(r, \theta, t) \cos(\theta) + b_2(r, \theta, \phi) + \cos(\theta) \left(h_2(\theta) \kappa \right. \right. \\
 \left. \left. + 1 \right) \sin(\theta) \omega_3(t)^2 \right) \\
 - \frac{u_2(0, \theta, t) p_{1,1}(0, \theta, t)}{\sqrt{u_2(0, \theta, t)^2 + u_3(0, \theta, t)^2}} \\
 0
 \end{aligned} \tag{22}$$

Angular momentum balance:

$simplify(Determinant(e));$

$$\frac{\left(R_E + h_2(\theta) \right) \sin(\theta) \left(\left(\frac{d}{d\theta} h_2(\theta) \right)^2 + R_E \left(R_E + h_2(\theta) \right) \right)}{R_E} \tag{23}$$

$Pos_vec := Vector([X[1], X[2], X[3]]);$

$$\begin{aligned}
 Pos_vec := & \left[\left[\left(R_E + h_2(\theta) + r \right) \cos(\phi) \sin(\theta) \right. \right. \\
 & \left. \left. - \frac{r \left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) \cos(\phi)}{R_E} \right] \right. \\
 & \left[\left(R_E + h_2(\theta) + r \right) \sin(\phi) \sin(\theta) \right.
 \end{aligned} \tag{24}$$

$$- \frac{r \left(\frac{d}{d\theta} h_2(\theta) \right) \cos(\theta) \sin(\phi)}{R_E} \Bigg],$$

$$\left[(R_E + h_2(\theta) + r) \cos(\theta) + \frac{r \left(\frac{d}{d\theta} h_2(\theta) \right) \sin(\theta)}{R_E} \right]$$

Inertia := *Matrix*(1..3, 1..3) :

Inertia1 := *simplify*(*add*(*Pos_vec*[*i*].*Pos_vec*[*i*], *i* = 1..3))
IdentityMatrix(3) :

Inertia2 := -*simplify*(*OuterProductMatrix*(*Pos_vec*,
Pos_vec)) :

Inertia := *simplify*((*Inertia1* + *Inertia2*) . *sqrt*(*det_M*)) :

for *i* **from** 1 **to** 3 **do**:

for *j* **from** 1 **to** 3 **do**:

Inertia[*i*, *j*] := *simplify*(*convert*(*series*(*simplify*(*int*(*Inertia*[*i*,
j], [*x*[3] = 0..2·Pi, *x*[1] = 0..*h*]), *assume* = [*R_E* > 0,
sin(*x*[2]) > 0, 2 *h*₂(*θ*) + *R_E* > 0]), *h*, 2), *polynom*));

end do;

end do;

Inertia;

$$\left[\left[\pi (R_E + h_2(\theta))^2 R_E (\cos(\theta)^2 + 1) (2 h_2(\theta) \right. \right. \quad (25)$$

$$\left. + R_E) \sin(\theta) h, 0, 0 \right],$$

$$\left[0, \pi (R_E + h_2(\theta))^2 R_E (\cos(\theta)^2 + 1) (2 h_2(\theta) \right. \\ \left. + R_E) \sin(\theta) h, 0 \right],$$

$$\left[0, 0, 2 \pi (2 h_2(\theta) + R_E) (R_E + h_2(\theta))^2 R_E \sin(\theta)^3 h \right]$$

g[1] := *Vector*(*diff*(*Pos_vec*, *x*[1])) : *g*[2] := *diff*(*Pos_vec*,
x[2]) : *g*[3] := *diff*(*Pos_vec*, *x*[3]) :

$$\begin{aligned}
h_rel &:= \text{simplify} \left(\text{CrossProduct} \left(\text{Pos_vec}, U[1] \right. \right. \\
&\quad \cdot \left(\frac{e[1]}{\sqrt{M[1, 1]}} \right) \left. \right) + \text{CrossProduct} \left(\text{Pos_vec}, u[2](x[2], t) \right. \\
&\quad \cdot \left(\frac{e[2]}{\sqrt{M[2, 2]}} \right) \left. \right) + \text{CrossProduct} \left(\text{Pos_vec}, u[3](x[2], t) \right. \\
&\quad \cdot \left(\frac{e[3]}{\sqrt{M[3, 3]}} \right) \left. \right) : \\
\textbf{for } i \textbf{ from } 1 \textbf{ to } 3 \textbf{ do:} \\
h_rel[i] &:= \text{simplify}(\text{int}(h_rel[i], [x[3] = 0..2 \cdot \text{Pi}, x[1] = 0..h]) \\
&\quad \cdot \sqrt{\det_M}, \text{assume} = [\sqrt{\det_M} > 0, R_E > 0, \sin(x[2]) \\
&\quad > 0, 2 h_2(\theta) + R_E > 0]) : \\
h_rel[i] &:= \text{simplify}(\text{convert}(\text{series}(h_rel[i], h, 2), \text{polynom})); \\
\textbf{end do:} \\
h_rel & \\
& \left[\begin{aligned} &0, \\ &0, \\ &2 u_3(\theta, t) \sqrt{R_E} \sin(\theta)^2 (R_E \\ &+ h_2(\theta))^2 \sqrt{2 h_2(\theta) + R_E} \pi h \end{aligned} \right]
\end{aligned} \tag{26}$$

$$\begin{aligned}
CB_Pos_vec &:= \text{subs}(R_E = r, \text{simplify}(\text{subs}(\{h[2](x[2]) = 0, r \\
&= 0\}, \text{Pos_vec}))); \\
CB_Pos_vec &:= \begin{bmatrix} r \cos(\phi) \sin(\theta) \\ r \sin(\phi) \sin(\theta) \\ r \cos(\theta) \end{bmatrix}
\end{aligned} \tag{27}$$

$$\begin{aligned}
CB_Inertia &:= \text{Matrix}(1..3, 1..3) : \\
CB_Inertia1 &:= \text{simplify}(\text{add}(CB_Pos_vec[i] \cdot CB_Pos_vec[i], i \\
&= 1..3)) \cdot \text{IdentityMatrix}(3) : \\
CB_Inertia2 &:= -\text{simplify}(\text{OuterProductMatrix}(CB_Pos_vec,
\end{aligned}$$

```

    CB_Pos_vec)) :
    CB_Inertia := simplify((CB_Inertia1 + CB_Inertia2)·x[1]2
    ·sin(x[2])) :
for i from 1 to 3 do:
for j from 1 to 3 do:
    CB_Inertia[i, j] :=
        simplify(convert(series(simplify(int(CB_Inertia[i, j], [x[3]
            = 0 ..2·Pi, x[1] = 0 ..R_0 + h[2](x[2]))), assume = [R_E > 0,
            sin(x[2]) > 0, 2 h2(θ) + R_E > 0]), h, 2), polynom));
end do;
end do;
    CB_Inertia;

```

$$\left[\left[\frac{(R_0 + h_2(\theta))^5 \pi (-\sin(\theta)^3 + 2 \sin(\theta))}{5}, 0, 0 \right], \right. \tag{28}$$

$$\left[0, \frac{(R_0 + h_2(\theta))^5 \pi (-\sin(\theta)^3 + 2 \sin(\theta))}{5}, 0 \right],$$

$$\left[0, 0, \frac{2 \pi \sin(\theta)^3 (R_0 + h_2(\theta))^5}{5} \right] \Bigg]$$

```

    int(sin(x[2])3, x[2] = 0 ..Pi);

```

$$\frac{4}{3} \tag{29}$$